

Solutions to Problem Set 1, Second Half

180.301 Microeconomic Theory - Spring 2026

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1. $\frac{MP_L}{w} = \frac{24}{8} = 3$, $\frac{MP_K}{r} = \frac{32}{16} = 2$. $\frac{MP_L}{w} > \frac{MP_K}{r}$, the firm should change its output mix towards more labor and less capital.

2. $MP_L = 2K^{1/3}L^{-1/3}$, $MP_K = K^{-2/3}L^{2/3}$. Optimal production has $MP_L/MP_K = w/r \Rightarrow 2K/L = 1/2 \Rightarrow L = 4K$.

Plugging this into the cost equation: $200L + 400K = 12000 \Rightarrow 800K + 400K = 12000 \Rightarrow K = 10, L = 40$

3. $MP_L = 0.5(0.5K^{1/3} + 0.5L^{1/3})^2 L^{-2/3}$, $MP_K = 0.5(0.5K^{1/3} + 0.5L^{1/3})^2 K^{-2/3}$.

Optimal production has $\frac{MP_L}{MP_K} = \frac{w}{r} \Rightarrow \left(\frac{K}{L}\right)^{2/3} = 4 \Rightarrow K = 8L$.

To produce 8 units of output, $Q = (0.5K^{1/3} + 0.5L^{1/3})^3 = (L^{1/3} + 0.5L^{1/3})^3 = 8 \Rightarrow L = \frac{64}{27}$, $K = \frac{512}{27}$

4. (a) True. Accounting cost of his time is zero, but economic cost is positive because his time has an opportunity cost.

(Alternative answer: False. Accounting cost of his time is zero, but the firm incurs other accounting costs such as rent, input costs, ...)

- (b) False. An unemployed worker's time has an opportunity cost, e.g. he could have spent the time on leisure activities, training, ...

5. (a) With K fixed at 4, $Q = KL + L = 4L + L = 100$ $L = 20$.

Total cost $TC = 100 \times 20 + 100 \times 4 = 2400$.

- (b) $MP_L = K + 1$, $MP_K = L$. Optimal production has $\frac{MP_L}{w} = \frac{MP_K}{r} \Rightarrow K = L - 1$.

To produce 100 units of output, $Q = KL + L = (L - 1)L + L = 100$

$\Rightarrow L = 10$, $K = 9$, Total cost $TC = 100 \times 10 + 100 \times 9 = 1900$.

6. (a) $P = MC \Rightarrow 100 = 4q \Rightarrow q = 25$

(b) Profit $= TR - TC = Pq - (200 + 2q^2) = 2500 - (200 + 2 \times 25^2) = 1050$.

- (c) In the short run: the minimum price with a positive output is achieved where $P = MC = AVC \Rightarrow P = 4q = 2q = 0$.

In the long run: the minimum price with a positive output is achieved where $P = MC = ATC \Rightarrow P = 4q = \frac{200}{q} + 2q$. Solving, we get $q = 10$, $P = 40$.